

Lecture N: Title of This Lecture

Your Name
abc12@yale.edu

BIS 527: Introduction to Health Data Science
Yale School of Public Health
Lecturer: Harsh Parikh • Fall 2026

TL;DR

Replace this text with your summary.

AI use declaration. Replace this text. Name the tool, say what it did, say what you verified and how, and say what you did not verify.

1 Motivation

2 Setup and Notation

Notation.

3 Main Development

4 Worked Example

5 Summary

Syntax examples (delete before submitting)

Emphasis and quotation

Use *emphasis* for a term you are introducing. Quotation marks are “two backticks and two apostrophes”. A number range uses two hyphens, as in 1908–1922.

Citing a source

Both commands pull from `scribe-refs.bib`. Use `\citep` when the citation sits in brackets at the end of a claim, and `\citet` when the author's name is part of your sentence.

Randomization separates an intervention from ordinary variation (Fisher, 1935). Tukey (1962) argued that data analysis extends beyond mathematical statistics.

To cite a source that is not yet in `scribe-refs.bib`, open that file and add an entry. Copy the shape of an entry that is already there. The key, the first thing inside the braces, is what you type inside `\citep{...}`.

Mathematics

Mathematics inside a sentence goes between dollar signs, like $n \log n$. Mathematics on its own line goes in an `equation` environment, which numbers it so you can refer back to it:

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i. \tag{1}$$

Refer to it with `\eqref`, as in Equation (1). The tilde in `Equation~\eqref` is a space that will never break across a line.

A definition, and referring back to it

Definition 1 (Sorted array). An array A of length n is *sorted* if $A[i] \leq A[i + 1]$ for every i from 1 to $n - 1$.

Definition 1 is what binary search requires. The pattern is always the same: put a `\label{...}` inside the thing you want to name, then write `\ref{...}` with the same text wherever you refer to it. LaTeX fills in the number, so it stays correct when you move things around. The same works for sections, tables, figures and equations.

A theorem and its proof

State the claim in a `theorem` environment and argue it in a `proof` environment. The proof ends with a hollow square, which LaTeX adds for you. Everything inside a theorem is set in italic by convention, so do not add italics yourself.

Theorem 1 (Cost of binary search). *Let A be a sorted array of $n \geq 1$ entries in the sense of Definition 1, and let t be the target value. Binary search on A executes the body of its loop at most $\lceil \log_2 n \rceil + 1$ times.*

Proof. Write m_k for the number of entries still under consideration after k iterations, so that $m_0 = n$. One iteration compares $A[mid]$ with t and then either returns or discards $A[mid]$ together with

everything on one side of it. Each of the two remaining pieces holds at most $\lfloor m_k/2 \rfloor$ entries, so

$$m_{k+1} \leq \left\lfloor \frac{m_k}{2} \right\rfloor \quad \text{and therefore} \quad m_k \leq \left\lfloor \frac{n}{2^k} \right\rfloor. \quad (2)$$

The loop runs only while at least one entry remains, so it stops at the first k with $\lfloor n/2^k \rfloor = 0$. That inequality holds exactly when $2^k > n$, which first occurs at $k = \lfloor \log_2 n \rfloor + 1$. The body therefore runs at most $\lfloor \log_2 n \rfloor + 1$ times. $\square$

A proof that ends on a displayed equation puts the square in the wrong place. Write `\qedhere` at the end of the display to fix it. A result you are quoting rather than proving needs no proof environment, but it does need a citation. `lemma`, `corollary` and `proposition` work the same way as `theorem` and share its numbering, so a lemma following Theorem 1 would be Lemma 2.

Lists

An unnumbered list:

- first point
- second point

A numbered list uses `enumerate` in place of `itemize`.

A table

Method	Comparisons	Time
Linear search	n	$O(n)$
Binary search	$\log_2 n$	$O(\log n)$

Table 1: Put the caption below a table. The `lrr` says there are three columns, the first left-aligned and the other two right-aligned.

A figure

Save your plot as a PDF or PNG in the same folder, then write this. The `width` setting scales the image to seven tenths of the text width.

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.7\textwidth]{myplot.pdf}
  \caption{Put the caption below a figure. Say what the reader should see.}
```

```

\label{fig:myplot}
\end{figure}

```

Draw figures in Python or with TikZ. Photographs of handwritten work are not accepted.

Python code

```

1 def binary_search(a, t):
2     lo, hi = 0, len(a) - 1
3     while lo <= hi:
4         mid = (lo + hi) // 2
5         if a[mid] == t:
6             return mid
7         if a[mid] < t:
8             lo = mid + 1
9         else:
10            hi = mid - 1
11    return None

```

Pseudocode

Tables, figures and pseudocode are *floats*. LaTeX moves them to wherever they fit best on the page, which may not be where you typed them. The [h] asks for "here if possible" and is only a request. Do not fight it, and never write "the table below". Refer to floats by number, as in Table 1.

Algorithm 1 Binary search

```

1:  $lo \leftarrow 1, hi \leftarrow n$ 
2: while  $lo \leq hi$  do
3:      $mid \leftarrow \lfloor (lo + hi)/2 \rfloor$ 
4:     if  $A[mid] = t$  then
5:         return  $mid$ 
6:     end if
7:     if  $A[mid] < t$  then
8:          $lo \leftarrow mid + 1$ 
9:     else
10:         $hi \leftarrow mid - 1$ 
11:    end if
12: end while
13: return NOT FOUND

```

References

Fisher, R. A. (1935). *The Design of Experiments*. Oliver and Boyd, Edinburgh.
 Tukey, J. W. (1962). The future of data analysis. *The Annals of Mathematical Statistics*, 33(1):1–67.